Lab + 实验报告

实验过程

这一小节中，所有 AI 帮助完成的部分都进行了标注

参考

- AI(DeepSeek)
- A RISC-V64 CPU using System Verilog

Step 0

参考 A RISC-V64 CPU using System Verilog，完成了乘除法器。

Step 1

扩展了相关指令，并添加了新的控制逻辑，使乘除法器计算过程中，整个流水线阻塞。

调试过程

这一小节中，所有 AI 帮助完成的部分都进行了标注

几乎没有什么太大问题。

lab1-extra

```
[src/cpu/cpu-exec.c:390,cpu_exec] nemu: HIT GOOD TRAP at pc = 0x000000008002001c
[src/cpu/cpu-exec.c:394,cpu_exec] trap code:0
[src/cpu/cpu-exec.c:74,monitor_statistic] host time spent = 1,982 us
[src/cpu/cpu-exec.c:76,monitor_statistic] total guest instructions = 32,775
[src/cpu/cpu-exec.c:77,monitor_statistic] simulation frequency = 16,536,326 instr/s
```

lab3-extra

```
[TEST] AES-128 test vector Encrypt
[PASS] AES-128 test vector Encrypt passed
[TEST] AES-128 test vector Decrypt
[PASS] AES-128 test vector Decrypt passed
[PASS] AES correctness checks passed
[RNG] heavy_rng_mod result=0, time(us)=43470
[RNG] heavy_rng_mod it/s=23255
[src/cpu/cpu-exec.c:390,cpu_exec] nemu: HIT GOOD TRAP at pc = 0x0000000080000030
[src/cpu/cpu-exec.c:394,cpu_exec] trap code:0
[src/cpu/cpu-exec.c:74,monitor_statistic] host time spent = 10,681 us
[src/cpu/cpu-exec.c:76,monitor_statistic] total guest instructions = 180,391
[src/cpu/cpu-exec.c:77,monitor_statistic] simulation frequency = 16,888,961 instr/s
```